

[Vitamin B6: Fact Sheet for Consumers \(original version\)](#)

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What is vitamin B6 and what does it do?

Vitamin B6 is a vitamin that is naturally present in many foods. The body needs vitamin B6 for more than 100 enzyme reactions involved in metabolism. Vitamin B6 is also involved in brain development during pregnancy and infancy as well as immune function.

How much vitamin B6 do I need?

The amount of vitamin B6 you need depends on your age. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	0.1 mg
Infants 7–12 months	0.3 mg
Children 1–3 years	0.5 mg
Children 4–8 years	0.6 mg
Children 9–13 years	1.0 mg
Teens 14–18 years (boys)	1.3 mg
Teens 14–18 years (girls)	1.2 mg
Adults 19–50 years	1.3 mg
Adults 51+ years (men)	1.7 mg
Adults 51+ years (women)	1.5 mg
Pregnant teens and women	1.9 mg
Breastfeeding teens and women	2.0 mg

What foods provide vitamin B6?

Vitamin B6 is found naturally in many foods and is added to other foods. You can get recommended amounts of vitamin B6 by eating a variety of foods, including the following:

- Poultry, fish, and organ meats, which are all rich in vitamin B6
- Potatoes and other starchy vegetables, which are some of the major sources of vitamin B6 for Americans
- Fruit (other than citrus), which are also among the major sources of vitamin B6 for Americans

What kinds of vitamin B6 dietary supplements are available?

Vitamin B6 is available in dietary supplements, usually in the form of pyridoxine. Most multivitamin/mineral supplements contain vitamin B6. Dietary supplements that contain only vitamin B6, or vitamin B6 with other B vitamins, are also available.

Am I getting enough vitamin B6?

Most people in the United States get enough vitamin B6 from the foods they eat. However, certain groups of people are more likely than others to have trouble getting enough vitamin B6:

- People whose kidneys do not work properly, including people who are on kidney dialysis and those who have had a kidney transplant
- People with autoimmune disorders, which cause their immune system to mistakenly attack their own healthy tissues, for example, people with rheumatoid arthritis, celiac disease, Crohn's disease, ulcerative colitis, or inflammatory bowel disease
- People with alcohol dependence

What happens if I don't get enough vitamin B6?

Vitamin B6 deficiency is uncommon in the United States. People who don't get enough vitamin B6 can have a range of symptoms, including anemia, itchy rashes, scaly skin on the lips, cracks at the corners of the mouth, and a swollen tongue. Other symptoms of very low vitamin B6 levels include depression, confusion, and a weak immune system. Infants who do not get enough vitamin B6 can become irritable or develop extremely sensitive hearing or seizures.

What are some effects of vitamin B6 on health?

Scientists are studying vitamin B6 to understand how it affects health. Here are some examples of what this research has shown.

Cardiovascular disease

Some scientists had thought that certain B vitamins (such as folic acid, vitamin B12, and vitamin B6) might reduce cardiovascular disease risk by lowering levels of homocysteine, an amino acid in the blood. Although vitamin B supplements do lower blood homocysteine, research shows that they do not actually reduce the risk or severity of heart disease or stroke.

Cancer

People with low levels of vitamin B6 in the blood might have a higher risk of certain kinds of cancer, such as colorectal cancer. However, studies to date have not shown that vitamin B6 supplements can help prevent cancer or lower the chances of dying from this disease.

Cognitive function

Some research indicates that elderly people who have higher blood levels of vitamin B6 have better memory. However, taking vitamin B6 supplements (alone or combined with vitamin B12 and/or folic acid) does not seem to improve cognitive function or mood in healthy people or in people with dementia.

Premenstrual syndrome

Scientists aren't yet certain about the potential benefits of taking vitamin B6 for premenstrual syndrome (PMS). However, some studies show that vitamin B6 supplements could reduce PMS symptoms, including moodiness, irritability, forgetfulness, bloating, and anxiety.

Nausea and vomiting in pregnancy

At least half of all women experience nausea, vomiting, or both in the first few months of pregnancy. Based on the results of several studies, the American College of Obstetricians and Gynecologists recommends taking vitamin B6 supplements under a doctor's care for nausea and vomiting during pregnancy.

Can vitamin B6 be harmful?

People almost never get too much vitamin B6 from food or beverages. However, taking high amounts of vitamin B6 from supplements for a year or longer can cause severe nerve damage, leading people to lose control of their bodily movements. The symptoms usually stop when they stop taking the supplements. Other symptoms of too much vitamin B6 include painful, unsightly skin patches, extreme sensitivity to sunlight, nausea, and heartburn.

The daily upper limits for vitamin B6 include intakes from all sources—food, beverages, and supplements—and are listed below. These levels do not apply to people who are taking vitamin B6 for medical reasons under the care of a doctor.

Life Stage	Upper Limit
Birth to 12 months	Not established
Children 1–3 years	30 mg
Children 4–8 years	40 mg
Children 9–13 years	60 mg
Teens 14–18 years	80 mg
Adults	100 mg

Some experts recommend lower limits for vitamin B6. For example, in 2023 the European Food Safety Authority set an upper limit for vitamin B6 of 12 mg per day for all adults, with lower amounts ranging from 2.2 to 10.7 mg per day for infants and children, depending on age.

Does vitamin B6 interact with medications or other dietary supplements?

Yes, vitamin B6 supplements can interact or interfere with medicines that you take. Here are several examples.

- Vitamin B6 supplements might interact with cycloserine (Seromycin), an antibiotic used to treat tuberculosis, and worsen any seizures and nerve cell damage that the drug might cause.
- Taking certain epilepsy drugs could decrease vitamin B6 levels and reduce the drugs' ability to control seizures.
- Taking theophylline (Aquaphyllin, Elixophyllin, Theolair, Truxophyllin, and many others) for asthma or another lung disease can reduce vitamin B6 levels and cause seizures.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and medicines you take. They can tell you if those dietary supplements might interact or interfere with your prescription or over-the-counter medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Vitamin B6 and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about vitamin B6?

- For general information on vitamin B6
 - [Office of Dietary Supplements \(ODS\) Health Professional Fact Sheet on Vitamin B6](#)
 - [Vitamin B6](#), MedlinePlus
- For more information on food sources of vitamin B6
 - USDA's [FoodData Central](#)
 - Nutrient List for vitamin B6 (listed by [food](#) or by [vitamin B6 content](#)), USDA
- For more advice on choosing dietary supplements
 - [ODS Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

alcohol dependence

A chronic disease (it lasts a person's lifetime) in which a person is unable to stop drinking once he or she has begun, needs to drink larger amounts of alcohol to get high, and suffers withdrawal symptoms (such as nausea, sweating, shakiness, and anxiety) after stopping drinking. The risk of developing alcohol dependence is influenced by a person's genes and lifestyle. Also called alcoholism.

amino acid

A chemical building block of protein.

anemia

A condition in which the number of red blood cells in the blood, or the amount of hemoglobin in them, is lower than normal, causing a condition in which red blood cells are not able to supply enough oxygen to all the tissues in the body. Hemoglobin is the substance in red blood cells that carries oxygen to your body's cells.

antibiotic

A drug used to treat infections caused by bacteria and other microorganisms.

asthma

A long-lasting disease in which the airways that carry air in and out of the lungs sometimes become narrower, limiting the amount of air that can flow through them, and causing wheezing, coughing, chest tightness, and trouble breathing (called a spasm). Cells in the airways may also make more mucus than usual, which narrows the airways even more. Asthma can also cause death, but can usually be controlled with regular treatment. It may be caused by allergies (such as to pet hair, dust, mold, pollen, or cockroaches), respiratory infections, vigorous exercise, environmental factors (such as cigarette smoke, certain foods, or pollution), stress, and genetics (a child is more likely to have asthma if one or both parents have asthma).

B vitamin

A nutrient that is important for cell function. The B vitamins are biotin, folate, niacin, pantothenic acid, riboflavin (vitamin B₂), thiamin, vitamin B₆, and vitamin B₁₂. The B vitamins make up the vitamin B complex.

cancer

A group of diseases in which cells divide abnormally and without control, and spread to nearby tissues and other parts of the body. Without treatment, cancer can stop organs from working normally, damage body systems, and cause the patient to die. Cancer may be caused by multiple factors, such as radiation, sunlight, tobacco, certain viruses, and poisonous chemicals; however, the cause of many cancers is unknown.

cardiovascular disease

CVD. A general term referring to disorders of the heart and blood vessels. CVD includes coronary artery disease, heart failure, atherosclerosis, high blood pressure, peripheral artery disease, and stroke.

celiac disease

An autoimmune disorder in which eating gluten (a protein found in wheat, rye, barley, and possibly oats) causes the immune system to damage the small intestine, making it unable to absorb nutrients. It is a genetic disease that sometimes becomes active for the first time after surgery, pregnancy, childbirth, viral infection, or extreme stress. Also called sprue.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

cognitive function

Mental awareness and judgment.

colorectal cancer

Uncontrolled growth of abnormal cells in the colon (the longest part of the large intestine) and/or the rectum (the last several inches of the large intestine before the anus).

Crohn's disease

A long-lasting (chronic) disease that causes severe irritation in the gastrointestinal tract. It usually affects the lower small intestine (called the ileum) or the colon, but it can affect any part of the digestive tract from the mouth to the anus. It is painful, causing severe watery or bloody diarrhea, and may lead to life-threatening complications. Crohn's disease is a form of inflammatory bowel disease.

deficiency

An amount that is not enough; a shortage.

dementia

Damaged brain function (thinking, learning, making decisions, remembering) that worsens over time. It disrupts activities of daily living such as bathing, dressing, and walking.

depression

A serious medical condition that can interfere with an individual's ability to work, study, sleep, and eat. Symptoms include ongoing feelings of sadness and despair, loss of energy, and loss of interest or pleasure in activities that were once enjoyed. A person who has depression may experience changes in eating or sleeping habits, and have thoughts of death or suicide.

dialysis

The process of filtering the blood when the kidneys are not able to cleanse it.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

enzyme

A protein that speeds up chemical reactions in the body.

epilepsy

A group of disorders that sometimes disrupts proper communication between brain cells, causing a seizure (a sudden change in behavior due to excessive electrical activity in the brain). It usually occurs in young children and the elderly. Epilepsy can be caused by abnormal brain development, brain damage, illness, tumors, or strokes. Often the cause is not known.

folic acid

The form of folate (a B vitamin occurring naturally in food) that is manufactured and used in supplements and fortified foods.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

homocysteine

An amino acid (a building block of protein). At high blood levels, it may increase the risk of developing coronary heart disease, heart attack, stroke, and Alzheimer's disease. Elevated homocysteine may also increase the risk of developing osteoporosis and bone fractures.

immune function

Substances made and action taken by cells that fight disease and infection.

immune system

A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant

A child younger than 12 months old.

inflammatory bowel disease

IBD. Long-lasting (chronic) problems that cause irritation and ulcers in the digestive tract. The most common disorders are ulcerative colitis and Crohn's disease.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

kidney

One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

lung

An organ in the chest that supplies oxygen to the body and removes carbon dioxide. There are two lungs in the body.

metabolism

All chemical changes that take place in a cell or an organism. These changes produce energy and basic materials needed for important life processes.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nerve

A bundle of microscopic fibers that carries messages back and forth from the brain to other parts of the body.

neuron

A nerve cell. Neurons send chemical and electrical messages throughout the nervous system that direct the body to function, move, think, and have emotions.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

poultry

Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

premenstrual syndrome (PMS)

Severe symptoms that occur 1 or 2 weeks before menstruation, including cramping, bloating, and tender breasts; food cravings; mood swings and irritability; and headache and fatigue.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

pyridoxine

A form of vitamin B6.

rheumatoid arthritis

An autoimmune disease that causes pain, swelling, and stiffness in the joints. It may cause severe joint damage, loss of function, and disability. The disease may last from months to a lifetime, and symptoms may improve and worsen over time.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

seizure

Sudden changes in behavior caused by excessive electrical activity in the brain.

stroke

A loss of blood flow to part of the brain. Strokes are caused by blood clots or broken blood vessels in the brain, and result in damage to a section of brain tissue. Symptoms include dizziness, numbness, weakness on one side of the body, and problems with talking or understanding language. The chance (risk) of stroke is increased by high blood pressure, older age, smoking, diabetes, high cholesterol, heart disease, a family history of stroke, and a build-up of fatty material inside the coronary arteries (atherosclerosis). See also NIH publication: Know Stroke. Know the Signs. Act in Time.

<http://www.ninds.nih.gov/disorders/stroke/knowstroke.htm>

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

tissue

A group or layer of cells in a living organism that work together to perform a specific function.

transplant

The replacement of tissue with tissue from the person's own body or from another person.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

tuberculosis

A disease caused by a specific type of bacteria that spreads from one person to another through the air. Tuberculosis can affect many parts of the body, but most often affects the lungs. A person may not have symptoms of tuberculosis for years, but they may appear when the patient becomes ill with a serious condition like diabetes, AIDS, or cancer. Tuberculosis can usually be treated and cured with antibiotics. Also called TB.

ulcerative colitis

Chronic inflammation of the colon that causes ulcers to form in its lining. This condition is marked by abdominal pain, cramps, and loose discharges of pus, blood, and mucus from the bowel.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin B₁₂

A group of chemical compounds that contain cobalt and are needed for certain chemical reactions in the body. Vitamin B₁₂ is involved in maintaining healthy nerve cells and red blood cells. It is needed to make DNA (the genetic material in all cells), and is required for the metabolism (chemical changes that take place in the tissues to produce energy and the basic materials needed by the body) of carbohydrate, fat, and protein. Also called cobalamin. For more information see the NIH Office of Dietary Supplements Vitamin B₁₂ fact sheet.

vitamin B₆

A group of water-soluble chemical compounds, including pyridoxine, pyridoxal, and pyridoxamine. It is involved in protein metabolism, is needed for the nervous system and immune system to work efficiently, and is needed to make hemoglobin (a molecule within red blood cells that carries oxygen to tissues). It also helps maintain blood glucose (sugar) within a normal range. For more information see the NIH Office of Dietary Supplements Vitamin B₆ fact sheet.

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